SEMESTER-VII

UEE708: POWER ELECTRONICS AND DRIVES				
	L	T	P	Cr
	3	0	2	4.0

Course Objective: To review the operational aspects of power electronic devices and principle of conversion and control of AC and DC voltages, concept of rotating machines, and drives.

Syllabus:

Introduction: Introduction to Thyristors and its family, turn-on and turn - off methods, BJT, MOSFET, IGBT, GTO, selection of devices for various applications.

Phase Controlled Converters: Principle of phase control, Single phase and three phase converter circuits with different types of loads.

AC Voltage Controllers: Types of single-phase voltage controllers, single-phase voltage controller with R and RL type of loads.

DC Choppers: Principle of chopper operation, types of choppers, step up and step down choppers.

Inverters: Single phase half bridge and full bridge inverter, three phase 120 and 180 conduction mode of inverters

Cycloconverters: Principles of operation, single phase to single phase step up and step down cycloconverters.

Electrical Machines: Principle and characteristics of DC machines, EMF and torque production of DC motor, four quadrant operation, speed control of DC motor, thyristor and chopper fed DC drives. Principle of operation of single phase and three phase Induction motor. Torque –speed and torque-slip characteristics, methods of starting of squirrel cage and slip ring motors.

Control Theory: Importance of Feedback control, requirement of feedback loops in drive applications, current-limit control and chopper fed dc motor drives.

Laboratory Work

SCR V-I characteristics, Gate firing circuit, DC -DC chopper, Semi converter and Full converter with R , RL and RLE type of loads, DC shunt motor speed control, Single phase .AC voltage controller with R load, Inverters, Simulation of power electronics converters

Course Learning Outcomes (CLOs)

The students will be able to:

1. exhibit the knowledge of various power semiconductor devices related to its characteristics
2. exhibit the knowledge of working principle and applications of phase controlled converters
3. explicate the working and applications DC choppers and cycloconverters
4. analyze and compare the performance of DC and AC machines in various drive applications
5. elucidate the concepts of feedback control theory

Text Books

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

1. Dubey, G.K., Doradla, S.R., Joshi, A. and Sinha, R.M.K., Thyristorised Power Controllers,New Age International (P) Limited, Publishers (2004).
2. Rashid, M., Power Electronics, Prentice Hall of India (2006).
3. Bimbhra,P.S., Power Electronics, Khanna Publishers(2012).
4. N. Mohan, T. M. Undeland, W.P Robbins, ―Power Electronics, Converters, Applications & Design, Wiley India Pvt. Ltd.
5. Daniel.W.Hart, "Power Electronics", Mc GrawHill Publications 2010.
6. Joseph Vithayathil, ―Power Electronics‖, Tata McGraw Hill.
7. Simon Ang, Alejandro Oliva, "Power-Switching Converters" Taylor and Francis group
8. Krishnan, R., Electric Motor Drives: Modeling, Analysis, and Control. Prentice Hall, (2001).

Reference Books

1. Mohan, N., Underland, T. and Robbins, W. P., Power Electronics: Converter Applications and Design, John Wiley (2007) 3rded.
2. Bose, B.K., Handbook of Power Electronics, IEEE Publications

Evaluation Scheme:

S N	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35